Step 2 : Data

Compute expected frequencies:

Under the assumption on H_0, the expected frequencies of the four grades are:

$$\frac{4}{10}\times 800 = 320\ ; \frac{3}{10}\times 800 = 240; \frac{2}{10}\times 800 = 160; \frac{1}{10}\times 800 = 80$$

Step 3 : Test statistic

The test statistic is computed using the following table.

Observed frequency (O_i)	Expected frequency (E_i)	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
320	320	0	0	0
270	240	30	900	3.75
190	160	30	900	5.625
20	80	-60	3600	45
			Total	54.375

The test statistic is calculated as

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$
$$= 54.375$$

Step 4 : Critical value

The critical value of χ^2 for 3 d.f. at 5% level of significance is 7.81 *i.e.*, $\chi^2_{3,0.05} = 7.81$

Step 5 : Decision

As the calculated value of χ_0^2(=54.375) is greater than the critical value $\chi^2_{3,0.05} = 7.81$, reject H_0. Hence, the results of the four grades do not follow the ratio 4:3:2:1.

Example 2.14

The following table shows the distribution of digits in numbers chosen at random from a telephone directory.

Digit	0	1	2	3	4	5	6	7	8	9
Frequency	1026	1107	997	966	1075	933	1107	972	964	853

Test whether the occurence of the digits in the directory are equal at 5% level of significance.

Step 1 : Null hypothesis H_0: The occurrence of the digits are equal in the directory.

Alternative hypothesis H_1: The occurrence of the digits are not equal in the directory.

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